习题 4 - 2 解答

1. 填括号, 使下列各等式成立：

（1）$\mathrm{d}x=(\quad)\mathrm{d}(7x-3)$；　　（2）$x^3\mathrm{d}x=(\quad)\mathrm{d}(3x^4-2)$；

（3）$x\mathrm{d}x=(\quad)\mathrm{d}(1-x^2)$；　　（4）$\dfrac{1}{1+9x^2}\mathrm{d}x=(\quad)\mathrm{d}(\arctan 3x)$；

（5）$\cos(\omega t+\varphi)\mathrm{d}t=\mathrm{d}(\quad)$；　　（6）$\mathrm{e}^{kx}\mathrm{d}x=\mathrm{d}(\quad)$；

（7）$\dfrac{x}{\sqrt{x^2+a^2}}\mathrm{d}x=\mathrm{d}(\quad)$；　　（8）$\sin x\cos x\mathrm{d}x=\mathrm{d}(\quad)$.

（9）$\sec^2 x\mathrm{d}x=(\quad)\mathrm{d}(2\tan x)$；　　（10）$\dfrac{1}{x}\mathrm{d}x=(\quad)\mathrm{d}(2-3\ln x)$.

解（1）因 $\mathrm{d}(7x-3)=7\mathrm{d}x$，故 $\dfrac{1}{7}\mathrm{d}(7x-3)=\mathrm{d}x$，所以 $\mathrm{d}x=\left(\dfrac{1}{7}\right)\mathrm{d}(7x-3)$.

（2）因 $\mathrm{d}(3x^4-2)=12x^3\mathrm{d}x$，故 $\dfrac{1}{12}\mathrm{d}(3x^4-2)=x^3\mathrm{d}x$，所以 $x^3\mathrm{d}x=\left(\dfrac{1}{12}\right)\mathrm{d}(3x^4-2)$.

（3）因 $\mathrm{d}(1-x^2)=-2x\mathrm{d}x$，故 $-\dfrac{1}{2}\mathrm{d}(1-x^2)=x\mathrm{d}x$，所以 $x\mathrm{d}x=\left(-\dfrac{1}{2}\right)\mathrm{d}(1-x^2)$.

（4）因 $\mathrm{d}(\arctan 3x)=\dfrac{3}{1+9x^2}\mathrm{d}x$，故 $\dfrac{1}{3}\mathrm{d}(\arctan 3x)=\dfrac{1}{1+9x^2}\mathrm{d}x$，所以 $\dfrac{1}{1+9x^2}\mathrm{d}x=\left(\dfrac{1}{3}\right)\mathrm{d}(\arctan 3x)$.

（5）因 $\left[\sin(\omega t+\varphi)\right]'=\cos(\omega t+\varphi)\cdot\omega$，即 $\mathrm{d}\left[\sin(\omega t+\varphi)\right]=\omega\cos(\omega t+\varphi)\mathrm{d}t$，故 $\dfrac{1}{\omega}\mathrm{d}\left[\sin(\omega t+\varphi)\right]=\cos(\omega t+\varphi)\mathrm{d}t$，亦即 $\mathrm{d}\left[\dfrac{1}{\omega}\sin(\omega t+\varphi)\right]=\cos(\omega t+\varphi)\mathrm{d}t$，所以 $\cos(\omega t+\varphi)\mathrm{d}t=\mathrm{d}\left[\dfrac{1}{\omega}\sin(\omega t+\varphi)\right]$.

（6）因 $\left(\mathrm{e}^{kx}\right)'=k\mathrm{e}^{kx}$，即 $\mathrm{d}\mathrm{e}^{kx}=k\mathrm{e}^{kx}\mathrm{d}x$，故 $\mathrm{d}\left(\dfrac{1}{k}\mathrm{e}^{kx}\right)=\mathrm{e}^{kx}\mathrm{d}x$，所以 $\mathrm{e}^{kx}\mathrm{d}x=\mathrm{d}\left(\dfrac{1}{k}\mathrm{e}^{kx}\right)$.

（7）因$\left(\sqrt{x^2+a^2}\right)'=\dfrac{2x}{2\sqrt{x^2+a^2}}=\dfrac{x}{\sqrt{x^2+a^2}}$，即$\mathrm{d}\sqrt{x^2+a^2}=\dfrac{x}{\sqrt{x^2+a^2}}\mathrm{d}x$，所以$\dfrac{x}{\sqrt{x^2+a^2}}\mathrm{d}x=\mathrm{d}\left(\sqrt{x^2+a^2}\right)$.

（8）因$\left(\sin^2 x\right)'=2\sin x\cos x$，即$\mathrm{d}\sin^2 x=2\sin x\cos x\mathrm{d}x$，故$\mathrm{d}\left(\dfrac{1}{2}\sin^2 x\right)=\sin x\cos x\mathrm{d}x$，所以$\sin x\cos x\mathrm{d}x=\mathrm{d}\left(\dfrac{1}{2}\sin^2 x\right)$.

（9）因$\mathrm{d}\left(2\tan x\right)=2\sec^2 x\mathrm{d}x$，故$\dfrac{1}{2}\mathrm{d}\left(2\tan x\right)=\sec^2 x\mathrm{d}x$，所以$\sec^2 x\mathrm{d}x=(\quad\dfrac{1}{2}\quad)\mathrm{d}\left(2\tan x\right)$;

（10）因$\mathrm{d}(2-3\ln x)=\left(-3\right)\dfrac{1}{x}\mathrm{d}x$，故$-\dfrac{1}{3}\mathrm{d}(2-3\ln x)=\dfrac{1}{x}\mathrm{d}x$，所以$\dfrac{1}{x}\mathrm{d}x=\left(-\dfrac{1}{3}\right)\mathrm{d}(2-3\ln x)$.

2. 设$\int f(x)dx=x^2+C$, 求$f(x)$, $\int xf(1-x^2)dx$.

解 对$\int f(x)dx=x^2+C$两边求导，得$f(x)=\left(\int f(x)dx\right)'=\left(x^2+C\right)'=2x$.

$$\int xf(1-x^2)dx=-\frac{1}{2}\int f(1-x^2)d(1-x^2)=-\frac{1}{2}(1-x^2)^2+C.$$

3. 设$f'(\cos x+2)=\sin^2 x+\tan^2 x$, 求$f(x)$.

解 对$f'(\cos x+2)=\sin^2 x+\tan^2 x$两边积分，得

$$\begin{aligned}f(\cos x+2)&=\int f'(\cos x+2)d(\cos x+2)=\int\left(\sin^2 x+\tan^2 x\right)d(\cos x+2)\\&=\int\left(1-\cos^2 x+\frac{1}{\cos^2 x}-1\right)d\cos x\\&=\int\left(-\cos^2 x+\frac{1}{\cos^2 x}\right)d\cos x\\&=-\frac{1}{3}\cos^3 x-\frac{1}{\cos x}+C.\end{aligned}$$

令$\cos x+2=t$，则$\cos x=t-2$，于是$f(t)=-\dfrac{1}{3}\left(t-2\right)^3-\dfrac{1}{t-2}+C$，即

$$f(x)=-\frac{1}{3}\left(x-2\right)^3-\frac{1}{x-2}+C.$$

4. 求下列不定积分.

(1) $\int x(1+x^2)^{100}dx;$

(2) $\int \sqrt{2+3x}dx;$

(3) $\int \cos(3x+2)dx;$

(4) $\int \frac{1}{\sqrt{1+\mathrm{e}^x}}dx;$

(5) $\int \frac{e^x}{1+e^x}dx;$

(6) $\int e^{e^x+x}dx;$

(7) $\int \frac{1}{e^x+e^{-x}}dx;$

(8) $\int \frac{\sin\left(\sqrt{t}+1\right)}{\sqrt{t}}dt;$

(9) $\int \frac{\sqrt{1-\sqrt{x}}}{\sqrt{x}}dx;$

(10) $\int \frac{1}{x\ln x\ln\ln x}dx;$

(11) $\int \frac{x}{\sqrt{1+x^2}}\tan\sqrt{1+x^2}dx;$

(12) $\int \frac{1}{x^2}\cos\frac{1}{x}dx;$

(13) $\int \frac{3x^3}{1-x^4}dx;$

(14) $\int \cos^2(\omega t+\varphi)dt;$

(15) $\int \frac{\sin x+\cos x}{\sqrt[3]{\sin x-\cos x}}dx;$

(16) $\int \cos^3 xdx;$

(17) $\int \cos 3x\sin 2xdx;$

(18) $\int \cos x\cos\frac{x}{2}dx;$

(19) $\int \frac{1}{\sin x\cos x}dx;$

(20) $\int \tan^3 x\sec xdx;$

(21) $\int \frac{1}{\cos^2 x\sqrt{1+\tan x}}dx;$

(22) $\int \frac{10^{2\arccos x}}{\sqrt{1-x^2}}dx;$

(23) $\int \frac{\arctan\sqrt{x}}{\sqrt{x}(1+x)}dx;$

(24) $\int \frac{1+\ln x}{(x\ln x)^2}dx;$

(25) $\int \frac{\ln\tan x}{\cos x\sin x}dx;$

(26) $\int \frac{1}{x\sqrt{x^2-1}}dx;$

(27) $\int \frac{\sqrt{x^2-9}}{x}dx;$

(28) $\int \frac{x^2}{\sqrt{a^2-x^2}}dx \quad (a>0);$

(29) $\int \frac{1}{1+\sqrt{2x}}dx;$

(30) $\int \frac{x}{1+\sqrt{1+x^2}}dx;$

(31) $\int \frac{1}{x^2\sqrt{a^2+x^2}}dx \quad (a>0);$ (32) $\int \frac{x^3}{(1+x^2)^{\frac{3}{2}}}dx;$

(33) $\int \frac{x^2}{(1+x^2)^2}dx;$ (34) $\int \frac{1}{x^2+2x+5}dx;$

(35) $\int \frac{1}{x(x^7+2)}dx;$ (36) $\int \frac{1}{(x+1)(x-2)}dx.$

解： (1) $\int x(1+x^2)^{100}dx=\frac{1}{2}\int(1+x^2)^{100}d(1+x^2)=\frac{1}{202}(1+x^2)^{101}+C.$

(2) $\int\sqrt{2+3x}dx=\frac{1}{3}\int(2+3x)^{\frac{1}{2}}d(2+3x)=\frac{2}{9}(2+3x)^{\frac{3}{2}}+C.$

(3) $\int\cos(3x+2)dx=\frac{1}{3}\int\cos(3x+2)d(3x+2)=\frac{1}{3}\sin(3x+2)+C.$

(4) $\int\frac{1}{\sqrt{1+\mathrm{e}^x}}dx\overset{\sqrt{1+\mathrm{e}^x}=t}{=\!=\!=}\int\frac{1}{t}d\ln(t^2-1)=\int\frac{2}{(t+1)(t-1)}dt=\int\frac{(t+1)-(t-1)}{(t+1)(t-1)}dt$

$=\int\left(\frac{1}{t-1}-\frac{1}{t+1}\right)dt=\ln(t-1)-\ln(t+1)+C=\ln\frac{t-1}{t+1}+C$

$=\ln\frac{\sqrt{1+\mathrm{e}^x}-1}{\sqrt{1+\mathrm{e}^x}+1}+C;$

(5) $\int\frac{e^x}{1+e^x}dx=\int\frac{1}{1+e^x}d(1+e^x)=\ln(1+e^x)+C;$

(6) $\int e^{e^x+x}dx=\int e^{e^x}\cdot e^x dx=\int e^{e^x}de^x=e^{e^x}+C;$

(7) $\int\frac{1}{\mathrm{e}^x+\mathrm{e}^{-x}}\mathrm{d}x=\int\frac{\mathrm{e}^x}{\mathrm{e}^{2x}+1}\mathrm{d}x=\int\frac{1}{(\mathrm{e}^x)^2+1}\mathrm{d}\mathrm{e}^x=\arctan\mathrm{e}^x+C.;$

(8) $\int\frac{\sin(\sqrt{t}+1)}{\sqrt{t}}dt=\int\sin(\sqrt{t}+1)\cdot\frac{1}{\sqrt{t}}\mathrm{d}t=2\int\sin(\sqrt{t}+1)\mathrm{d}\sqrt{t}$

$=2\int\sin(\sqrt{t}+1)\mathrm{d}(\sqrt{t}+1)=-2\cos(\sqrt{t}+1)+C;$

(9) $\int\frac{\sqrt{1-\sqrt{x}}}{\sqrt{x}}\mathrm{d}x=\int\sqrt{1-\sqrt{x}}\cdot\frac{1}{\sqrt{x}}\mathrm{d}x=2\int\sqrt{1-\sqrt{x}}\mathrm{d}\sqrt{x}$

$=-2\int\sqrt{1-\sqrt{x}}\mathrm{d}(1-\sqrt{x})=-\frac{4}{3}(1-\sqrt{x})^{\frac{3}{2}}+C.$

(10) $\int\frac{1}{x\ln x\ln\ln x}\mathrm{d}x=\int\frac{1}{\ln x\ln\ln x}\cdot\frac{1}{x}\mathrm{d}x=\int\frac{1}{\ln x\ln\ln x}\mathrm{d}\ln x$

$$= \int \frac{1}{\ln \ln x} \cdot \frac{1}{\ln x} \mathrm{d}\ln x = \int \frac{1}{\ln \ln x} \mathrm{d}\ln\ln x = \ln\left|\ln \ln x\right| + C.$$

(11) $$\int \frac{x}{\sqrt{1+x^2}} \tan\sqrt{1+x^2}\,\mathrm{d}x = \int \tan\sqrt{1+x^2}\,\frac{x}{\sqrt{1+x^2}}\,\mathrm{d}x$$

$$= \int \tan\sqrt{1+x^2}\,\mathrm{d}\sqrt{1+x^2} = -\ln\left|\cos\sqrt{1+x^2}\right| + C.$$

(12) $$\int \frac{1}{x^2}\cos\frac{1}{x}dx;\ \int \frac{1}{x^2}\cos\frac{1}{x}\mathrm{d}x = -\int \cos\frac{1}{x}\mathrm{d}\left(\frac{1}{x}\right) = -\sin\frac{1}{x} + C.$$

(13) $$\int \frac{3x^3}{1-x^4}dx = -\frac{3}{4}\int \frac{1}{1-x^4}d\left(1-x^4\right) = -\frac{3}{4}\ln\left|1-x^4\right| + C;$$ (1

（14） $$\int \cos^2(\omega t+\varphi)\mathrm{d}t = \int \frac{\cos 2(\omega t+\varphi)+1}{2}\mathrm{d}t = \frac{1}{2}\int \left(\cos 2(\omega t+\varphi)+1\right)\mathrm{d}t$$

$$= \frac{\sin 2(\omega t+\varphi)}{4\omega} + \frac{t}{2} + C.$$

(15) $$\int \frac{\sin x+\cos x}{\sqrt[3]{\sin x-\cos x}}\mathrm{d}x = \int \frac{1}{\sqrt[3]{\sin x-\cos x}}\left(\sin x+\cos x\right)\mathrm{d}x$$

$$= \int \left(\sin x-\cos x\right)^{-\frac{1}{3}}\mathrm{d}\left(\sin x-\cos x\right)$$

$$= \frac{3}{2}\left(\sin x-\cos x\right)^{\frac{2}{3}} + C.$$

(16) $$\int \cos^3 x\mathrm{d}x = \int \cos^2 x\cos x\mathrm{d}x = \int \left(1-\sin^2 x\right)\mathrm{d}\left(\sin x\right) = \sin x - \frac{1}{3}\sin^3 x + C.$$

(17) $$\int \cos 3x\sin 2x dx = \frac{1}{2}\int \left(\sin 5x-\sin x\right)\mathrm{d}x = -\frac{1}{10}\cos 5x + \frac{1}{2}\cos x + C.$$

(18) $$\int \cos x\cos\frac{x}{2}dx = \int \frac{1}{2}\left(\cos\frac{3x}{2}+\cos\frac{x}{2}\right)dx = \frac{1}{3}\sin\frac{3x}{2} + \sin\frac{x}{2} + C;$$

(19) $$\int \frac{1}{\sin x\cos x}\mathrm{d}x = 2\int \frac{1}{\sin 2x}\mathrm{d}x = \int \csc 2x\,\mathrm{d}\left(2x\right)$$

$$= \ln\left|\csc 2x-\cot 2x\right| + C = \ln\left|\tan x\right| + C$$

(20) $$\int \tan^3 x\sec x\mathrm{d}x = \int \tan^2 x\left(\tan x\sec x\right)\mathrm{d}x = \int \left(\sec^2 x-1\right)\mathrm{d}\left(\sec x\right)$$

$$= \frac{1}{3}\sec^3 x - \sec x + C.$$

(21) $$\int \frac{1}{\cos^2 x\sqrt{1+\tan x}}dx = \int \frac{1}{\sqrt{1+\tan x}}d\tan x = \int \frac{1}{\sqrt{1+\tan x}}d\left(1+\tan x\right)$$

$$= 2\sqrt{1+\tan x} + C;$$

(22) $\int \frac{10^{2\arccos x}}{\sqrt{1-x^2}}\mathrm{d}x = \int 10^{2\arccos x}\cdot\frac{1}{\sqrt{1-x^2}}\mathrm{d}x$

$$= -\frac{1}{2}\int 10^{2\arccos x}\mathrm{d}\left(2\arccos x\right) = -\frac{10^{2\arccos x}}{2\ln 10} + C.$$

(23) $\int \frac{\arctan\sqrt{x}}{\sqrt{x}(1+x)}\mathrm{d}x = \int \arctan\sqrt{x}\,\frac{1}{(1+x)}\cdot\frac{1}{\sqrt{x}}\mathrm{d}x = 2\int \arctan\sqrt{x}\,\frac{1}{(1+x)}\mathrm{d}\sqrt{x}$

$$= 2\int \arctan\sqrt{x}\cdot\frac{1}{1+\left(\sqrt{x}\right)^2}\mathrm{d}\sqrt{x}$$

$$= 2\int \arctan\sqrt{x}\mathrm{d}\left(\arctan\sqrt{x}\right) = \left(\arctan\sqrt{x}\right)^2 + C.$$

(24) $\int \frac{1+\ln x}{(x\ln x)^2}\mathrm{d}x = \int \frac{1}{(x\ln x)^2}\left(1+\ln x\right)\mathrm{d}x = \int \frac{1}{(x\ln x)^2}\mathrm{d}(x\ln x) = -\frac{1}{x\ln x} + C.$

(25) $\int \frac{\ln\tan x}{\cos x\sin x}dx = \int \frac{\cos x\cdot\ln\tan x}{\cos^2 x\sin x}dx = \int \frac{\ln\tan x}{\cos^2 x\tan x}dx = \int \frac{\ln\tan x}{\tan x}\sec^2 x dx$

$$= \int \frac{\ln\tan x}{\tan x}d\tan x = \int \ln\tan x d\ln\tan x = \frac{1}{2}\ln^2\tan x + C;$$

(26) 当 $x\in(1,+\infty)$ 时，令 $x=\sec t$，$t\in\left(0,\frac{\pi}{2}\right)$，则 $\sqrt{x^2-1}=\tan t$，

$\mathrm{d}x = \sec t\tan t\mathrm{d}t$，于是

$$\int \frac{\mathrm{d}x}{x\sqrt{x^2-1}} = \int \frac{\sec t\tan t}{\sec t\tan t}\mathrm{d}t = \int \mathrm{d}t = t + C = \arccos\frac{1}{x} + C.$$

(27) 令 $x=3\sec t$，则 $\sqrt{x^2-9}=3\tan t$，$\mathrm{d}x = 3\sec t\tan t\mathrm{d}t$，于是

$$\int \frac{\sqrt{x^2-9}}{x}\mathrm{d}x = \int \frac{3\tan t}{3\sec t}\cdot 3\sec t\tan t\mathrm{d}t = 3\int \tan^2 t\mathrm{d}t = 3\int(\sec^2 t-1)\mathrm{d}t = 3\tan t - 3t + C.$$

$$= 3\cdot\frac{\sqrt{x^2-9}}{3} - 3\arccos\frac{3}{x} + C = \sqrt{x^2-9} - 3\arccos\frac{3}{x} + C.$$

(28) 令 $x=a\sin t$，则 $\sqrt{a^2-x^2}=a\cos t$，$dx = a\cos t\mathrm{d}t$，于是

$$\int \frac{x^2}{\sqrt{a^2-x^2}}\mathrm{d}x = \int \frac{a^2\sin^2 t}{a\cos t}\cdot a\cos t\mathrm{d}t = \frac{a^2}{2}\int(1-\cos 2t)\mathrm{d}t = \frac{a^2}{2}t - \frac{a^2}{4}\sin 2t + C$$

$$= \frac{a^2}{2}t - \frac{a^2}{2}\sin t\cos t + C = \frac{a^2}{2}\arcsin\frac{x}{a} - \frac{a^2}{2}\cdot\frac{x}{a}\cdot\frac{\sqrt{a^2-x^2}}{a} + C$$

$$= \frac{a^2}{2}\arcsin\frac{x}{a} - \frac{x\sqrt{a^2-x^2}}{2} + C.$$

(29) $\sqrt{2x}=t$, 则$x=\frac{1}{2}t^2$, $\mathrm{d}x=t\mathrm{d}t$ 于是

$$\int\frac{\mathrm{d}x}{1+\sqrt{2x}}=\int\frac{t\mathrm{d}t}{1+t}=\int\frac{(t+1)-1}{1+t}\mathrm{d}t=\int\left(1-\frac{1}{1+t}\right)\mathrm{d}t$$

$$=t-\ln(t+1)+C=\sqrt{2x}-\ln(\sqrt{2x}+1)+C.$$

$$(30)\quad\int\frac{x}{1+\sqrt{1+x^2}}dx=\frac{1}{2}\int\frac{1}{1+\sqrt{1+x^2}}d\left(1+x^2\right)\overset{1+x^2=u}{=\!=}\frac{1}{2}\int\frac{1}{1+\sqrt{u}}du$$

$$\overset{\sqrt{u}=t}{=\!=}\frac{1}{2}\int\frac{1}{1+t}2tdt=\int\frac{t}{1+t}dt=\int\frac{(1+t)-1}{1+t}dt=\int\left(1-\frac{1}{1+t}\right)dt$$

$$=t-\ln\left|1+t\right|+C=\sqrt{1+x^2}-\ln\left|1+\sqrt{1+x^2}\right|+C.$$

$$(31)\quad\int\frac{1}{x^2\sqrt{a^2+x^2}}dx\overset{x=\frac{1}{t}}{=\!=}\int\frac{1}{\frac{1}{t^2}\sqrt{a^2+\frac{1}{t^2}}}\left(-\frac{1}{t^2}\right)dt=-\int\frac{t}{\sqrt{a^2t^2+1}}dt$$

$$=-\frac{1}{2a^2}\int\frac{1}{\sqrt{a^2t^2+1}}d\left(a^2t^2+1\right)=-\frac{1}{a^2}\sqrt{a^2t^2+1}+C$$

$$=-\frac{1}{a^2}\sqrt{a^2\frac{1}{x^2}+1}+C=-\frac{\sqrt{a^2+x^2}}{a^2x}+C.$$

(32) 令 $x=\tan t$， $\mathrm{d}x=\sec^2 t\mathrm{d}t$， $t=\arctan x$， 于是

$$\int\frac{x^3}{(1+x^2)^{\frac{3}{2}}}\mathrm{d}x=\int\frac{\tan^3 t}{\sec^3 t}\cdot\sec^2 t\mathrm{d}t=\int\frac{\sin^3 t}{\cos^2 t}\mathrm{d}t=-\int\frac{1-\cos^2 t}{\cos^2 t}\mathrm{d}(\cos t)$$

$$=\cos t+\frac{1}{\cos t}+C=\frac{1}{\sqrt{1+x^2}}+\sqrt{1+x^2}+C.$$

$$(33)\int\frac{x^2}{(1+x^2)^2}dx\overset{x=\frac{1}{t}}{=\!=}\int\frac{\frac{1}{t^2}}{\left(1+\frac{1}{t^2}\right)^2}\left(-\frac{1}{t^2}\right)dt=-\int\frac{1}{\left(t^2+1\right)^2}dt$$

$$\overset{t=\tan u}{=\!=}-\int\frac{1}{\sec^4 u}\sec^2 udu=-\int\cos^2 udu=-\int\frac{1+\cos 2u}{2}du$$

$$=-\frac{1}{2}\int(1+\cos 2u)du=-\frac{1}{2}u-\frac{1}{4}\sin 2u+C$$

$$=-\frac{1}{2}u-\frac{1}{2}\sin u\cos u+C=-\frac{1}{2}\arctan t-\frac{t}{2\left(1+t^2\right)}+C$$

$$=-\frac{1}{2}\arctan\frac{1}{x}-\frac{x}{2\left(1+x^2\right)}+C.$$

(34) 令 $x+1=2\tan t$，则 $\mathrm{d}x=2\sec^2 t\mathrm{d}t$，$t=\arctan\dfrac{x+1}{2}$，故

$$\int\frac{1}{x^2+2x+5}\mathrm{d}x=\int\frac{\mathrm{d}x}{4+(x+1)^2}=\int\frac{2\sec^2 t}{4\sec^2 t}\mathrm{d}t=\frac{1}{2}t+C=\frac{1}{2}\arctan\frac{x+1}{2}+C.$$

$$(35)\int\frac{1}{x(x^7+2)}dx=\int\frac{x^6}{x^7(x^7+2)}dx=\frac{1}{7}\int\frac{1}{x^7(x^7+2)}dx^7$$

$$=\frac{1}{14}\int\frac{\left(x^7+2\right)-x^7}{x^7(x^7+2)}dx^7=\frac{1}{14}\int\left(\frac{1}{x^7}-\frac{1}{(x^7+2)}\right)dx^7$$

$$=\frac{1}{14}\int\frac{1}{x^7}dx^7-\frac{1}{14}\int\frac{1}{(x^7+2)}d\left(x^7+2\right)$$

$$=\frac{1}{14}\ln\left|x^7\right|-\frac{1}{14}\ln\left|x^7+2\right|+C=\frac{1}{14}\ln\left|\frac{x^7}{x^7+2}\right|+C.$$

$$(36)\int\frac{1}{(x+1)(x-2)}dx=\frac{1}{3}\int\frac{(x+1)-(x-2)}{(x+1)(x-2)}dx=\frac{1}{3}\left[\int\frac{1}{(x-2)}-\frac{1}{(x+1)}\right]dx$$

$$=\frac{1}{3}\left(\ln\left|x-2\right|-\ln\left|x+1\right|\right)+C=\frac{1}{3}\left(\ln\frac{\left|x-2\right|}{\left|x+1\right|}\right)+C.$$